



Shrimps n Trees

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Introduction

- Create a tool that can be used to classify land-cover types, specifically mangrove vs non-mangrove
 - Compare over time to see change (deforestation or growth)
- Packages/Tools Used
 - numpy, pandas, os, time, datetime, matplotlib
 - xarray, rioxarray, rasterio, rasterdata, geopandas, osgeo
 - odc.stac, scipy, skimage, pystac_client, planetary_computer, xyzservices

Significance of Mangroves

Mangrove forests are a key ecosystem, containing thousands of species.

Stabilize coastlines from erosion.

Protect sea from pollution and land from typhoons.

35% of global mangrove cover lost in past 60 years- shrimp aquaculture, a major contributor to mangrove conversion.



Shrimp Farming in Vietnam

- Shrimp farming is a key source of food and income
- Shrimp pond aquaculture → negative impact on environment
- Importance of mangrove-integrated shrimp ponds
 - Visible with classification?
- Cycles of harvesting and planting mangrove trees
- Project focus: Ca Mau Peninsula, at the south end of Vietnam
 - Mangrove classification using raster data has been done for this region before



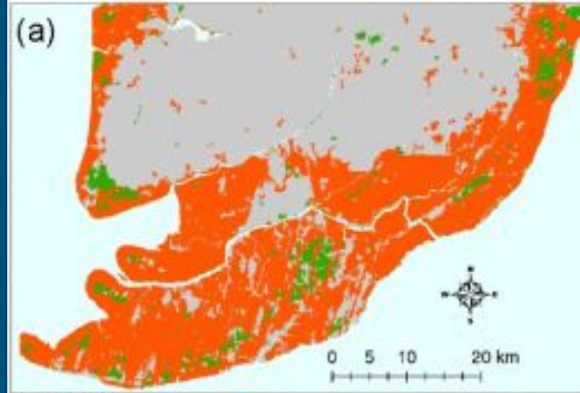
Mangrove Cover Over Time

(a) 1979

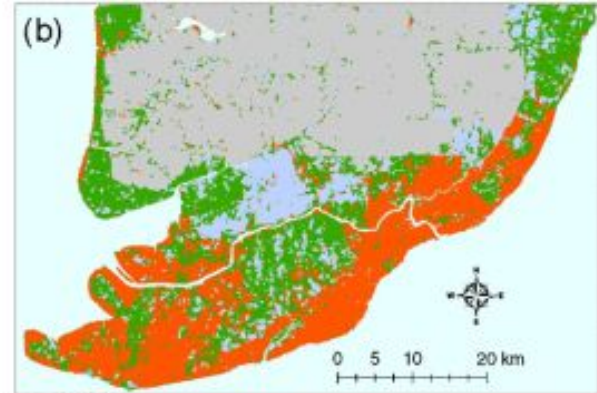
(b) 1989

(c) 2003

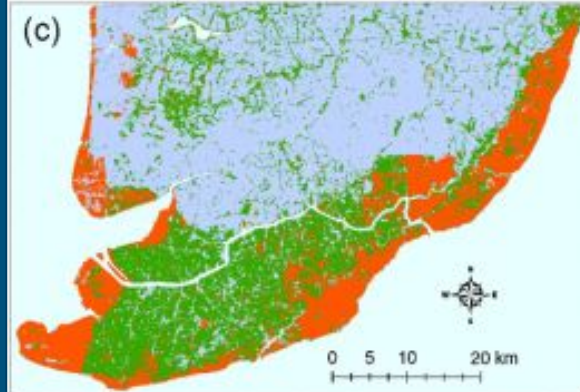
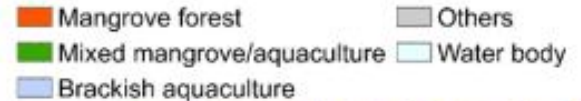
(d) 2013



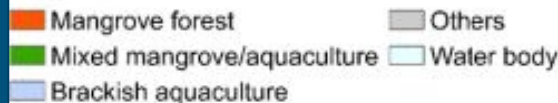
Legend



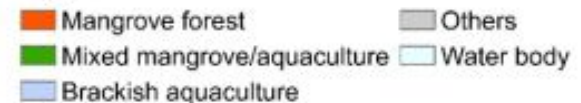
Legend



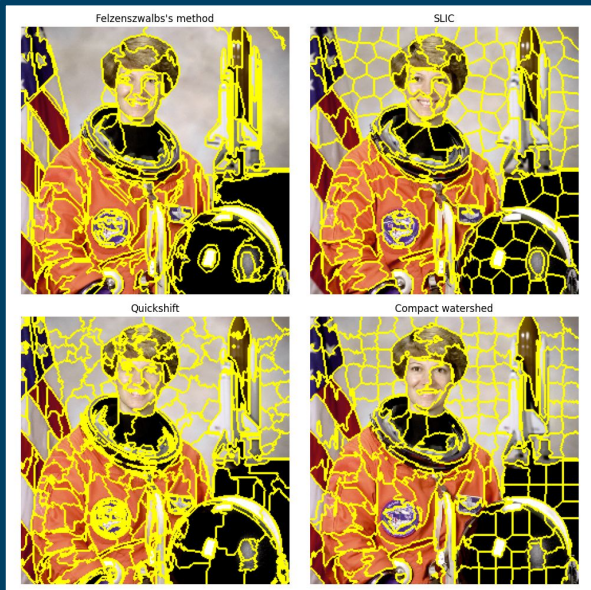
Legend



Legend



Object Based Image Classification

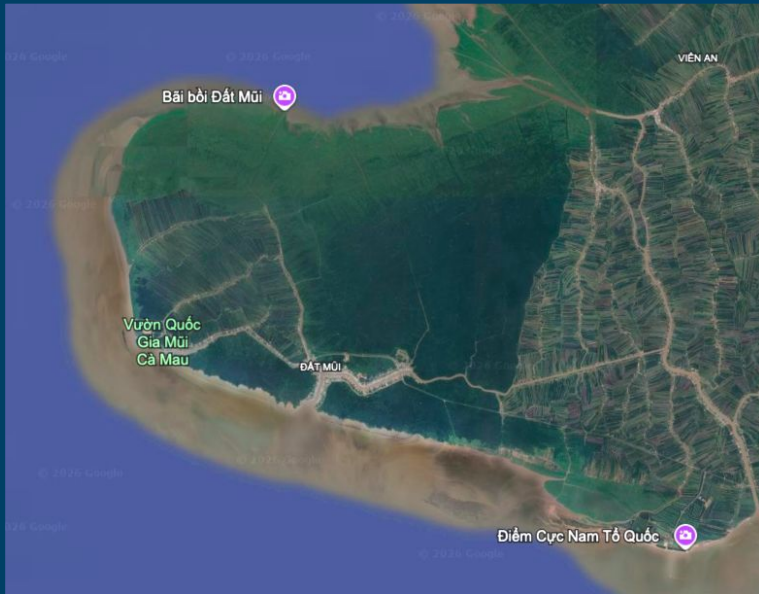


From scikit-image

Steps to complete...

- Image segmentation
 - Different algorithms for segmentation
 - Different parameters for segment shape/size
 - Classification from training data
 - Followed tutorial from OpenSourceOptions
-

Data used



Sentinel 2 datasets

Searched for data within set bounds of peninsula

Bands: G, R, NIR, SWIR1

NDVI, NDWI, CMRI (Combined Mangrove Recognition Index)

2017: oldest available data

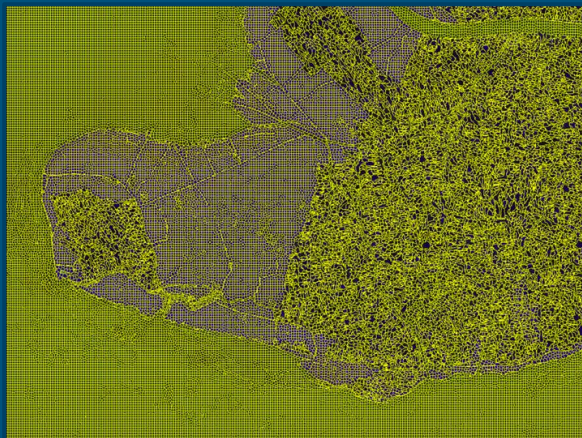
2026: newest available data

Selected months in dry season (Dec-April) and filtered to them

Filtered to days with low cloud cover

Harmonizing old and new datasets through normalization

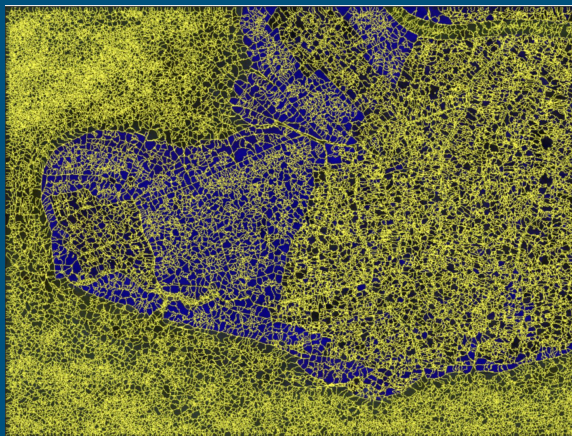
Method 1



2026 Segmentation Using slic:

n_segments=50000,
compactness=0.1

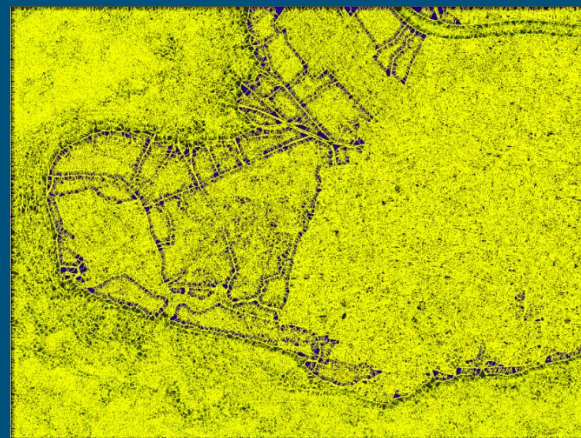
Method 2



2026 Segmentation Using
quickshift:

ratio=0.6, max_dist=4,
convert2lab=False

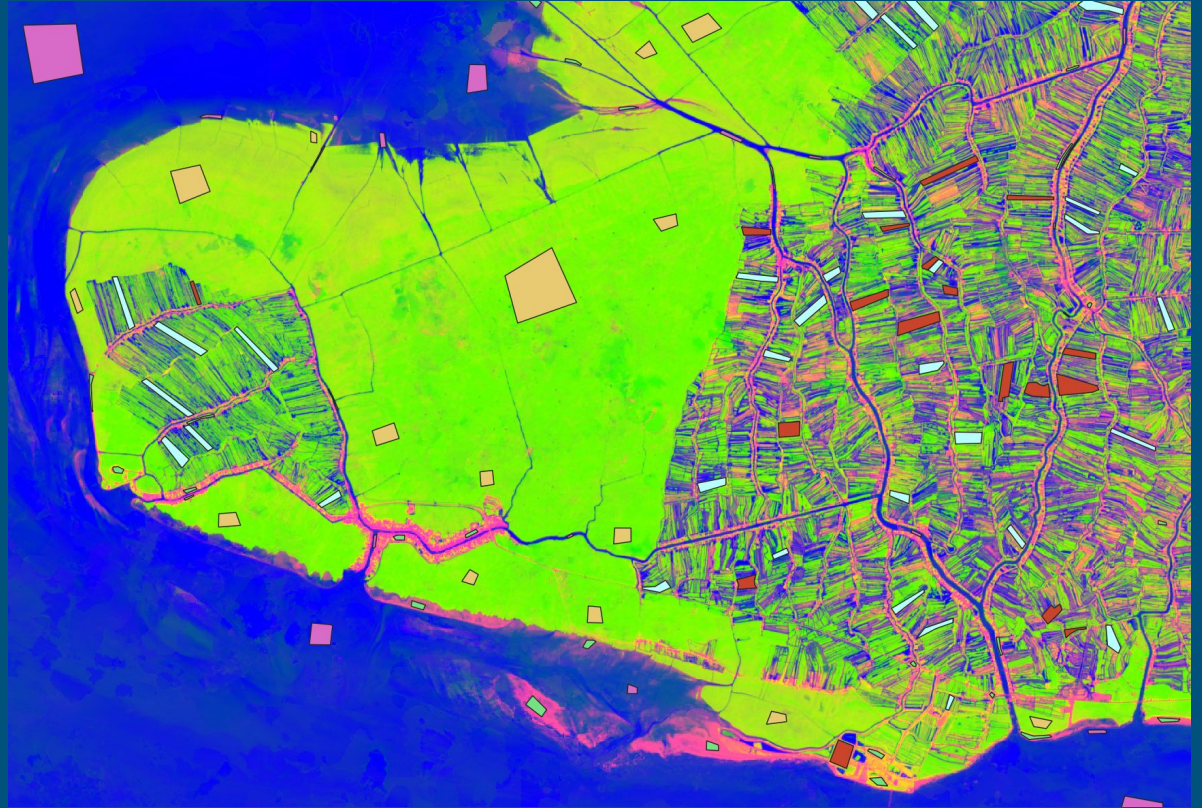
quickshift experimentation



2026 Segmentation Using
quickshift:

ratio=0.5, max_dist=2,
convert2lab=False

Training Data



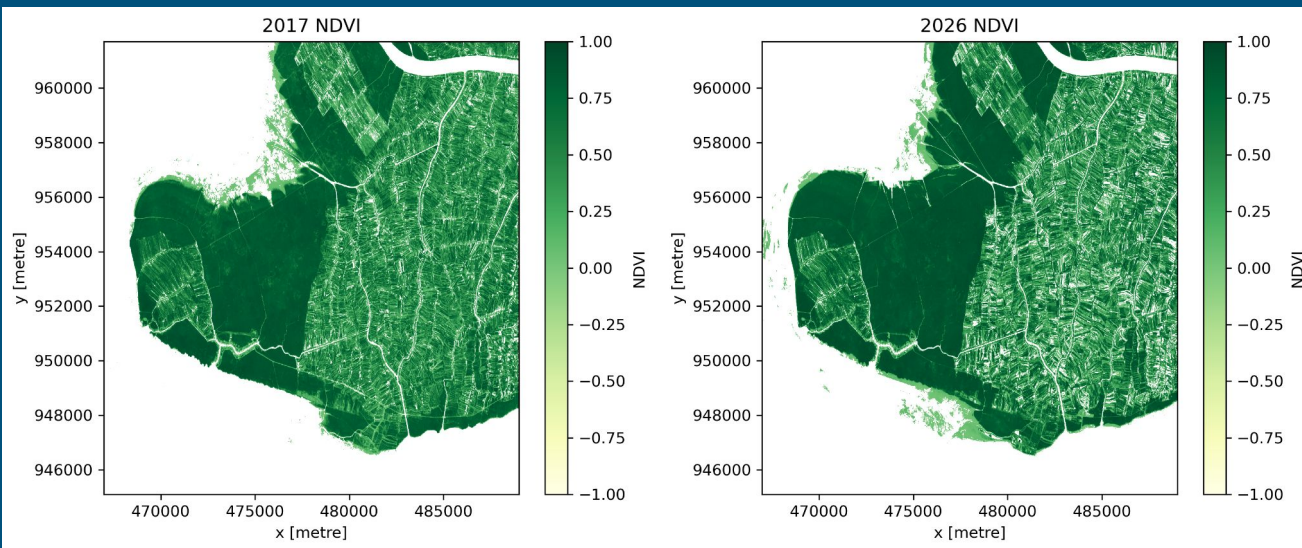
Threshold-Classification Analysis

- Similar to methods used in class
- NDVI (median across rasters) classification
- CMRI (median across rasters) classification
- Threshold classification (how we got there)
 - Mangrove age/ health
- Determined areas of mangrove vs non-mangrove
- Baseline of mangrove analysis

NDVI (Normalized Difference Vegetation Index)

$$\text{NDVI} = \text{NIR} - \text{RED} / \text{NIR} + \text{RED}$$

- Healthy vegetation reflects near infrared and absorbs red wavelength for photosynthesis.



NDVI can present issues with mangroves since they are often mixed with tidal flats, coastal vegetations and other water bodies.

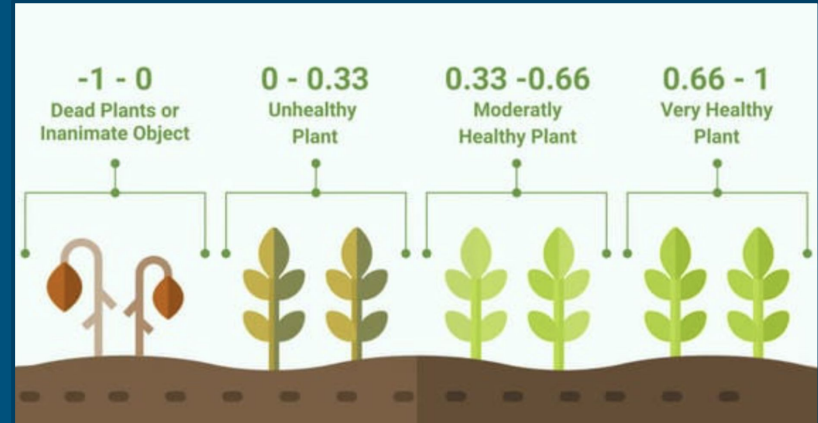
NDVI Thresholds/ Classification

Mean NDVI change: 0.0446

NDVI change std: 0.2481237

Threshold from ESRI

	Category	NDVI Threshold	2017 (%)	2026 (%)
0	Very Healthy	0.66 – 1.00	27.7	31.2
1	Healthy	0.33 – 0.66	17.3	14.0
2	Unhealthy	0.00 – 0.33	13.9	14.0



Sciencedirect threshold

	Category	NDVI Threshold	2017 (%)	2026 (%)
0	Healthy young / dense mature mangroves	0.40 – 0.68	15.2	12.2
1	Environmentally stressed mangroves	0.38 – 0.47	4.5	3.7

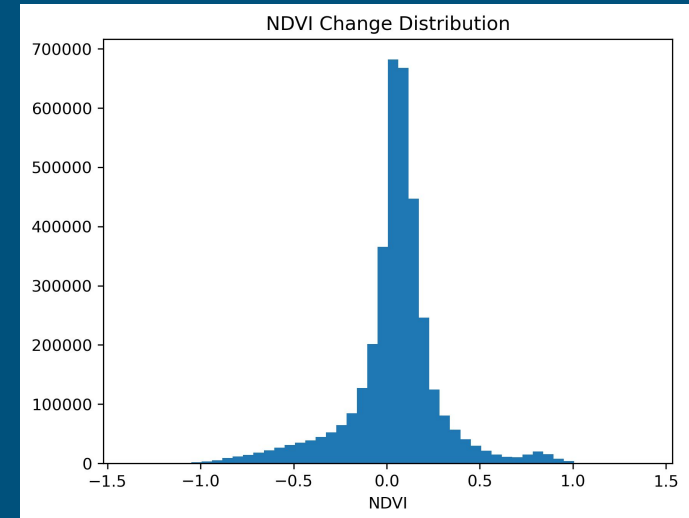
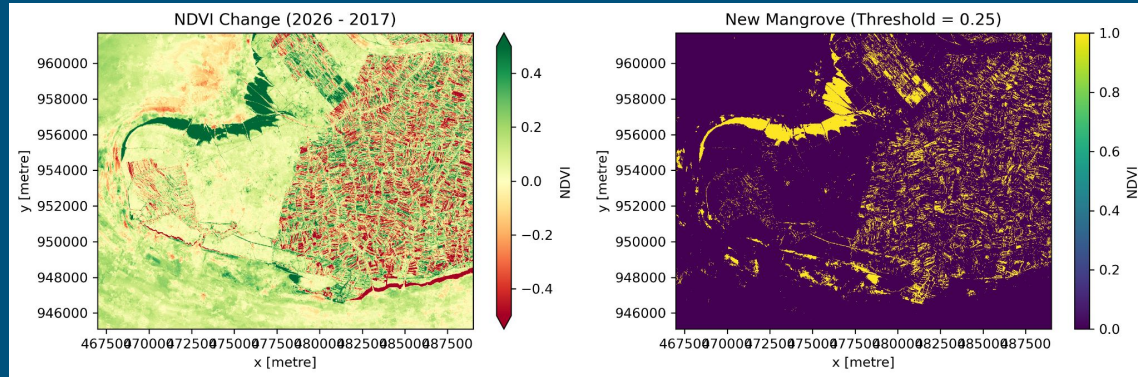
NDVI Threshold

Establishing threshold

- No ground truth
- Examples thresholds from literature
- Adapting these into visual analysis of plots and histogram

Classify density loss/ gain (0.1)

- Vegetation Gain: 34.17%
- Vegetation Loss: 19.55%
- Stable: 46.28%



Combined Mangrove Recognition Index (CMRI)

- CMRI excels at separating mangroves from water/ other coastal vegetation and aquatic environments. Mangroves are higher in chlorophyll and moisture than other plants. Chlorophyll absorbs red and blue, reflecting green band.
- CMRI produces better accuracy (73.43%). (Gupta k, et al)

$$\text{CMRI} = \frac{\text{Green} - \text{SWIR1}}{\text{NIR} + \text{SWIR1}}$$

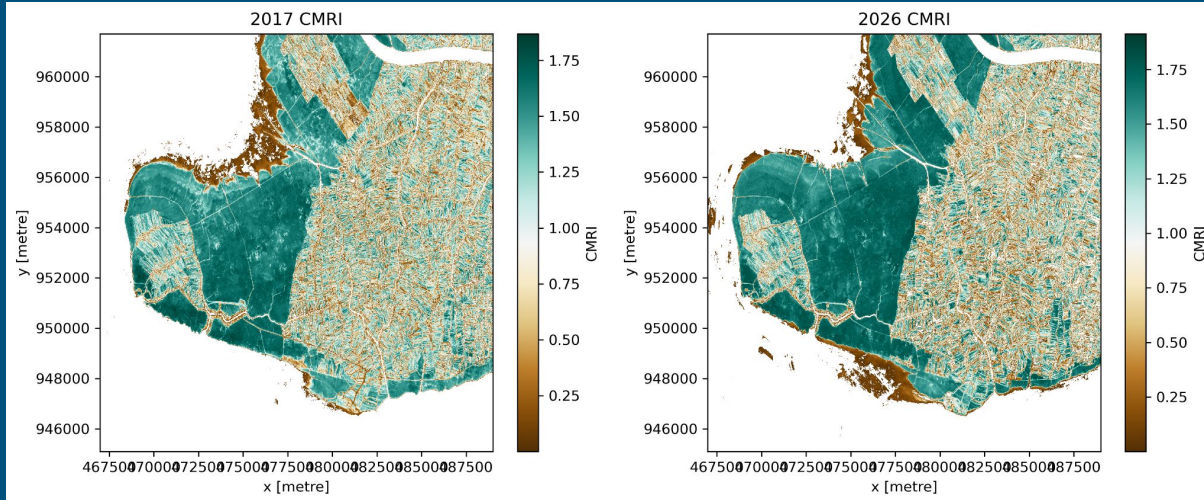
Using (Gupta k, et al) Mangrove forest index of 0.47-

2017 Mangrove Forests : 49.47%

2026 Mangrove Forests: 48.70%

Mean CMRI change: 0.07

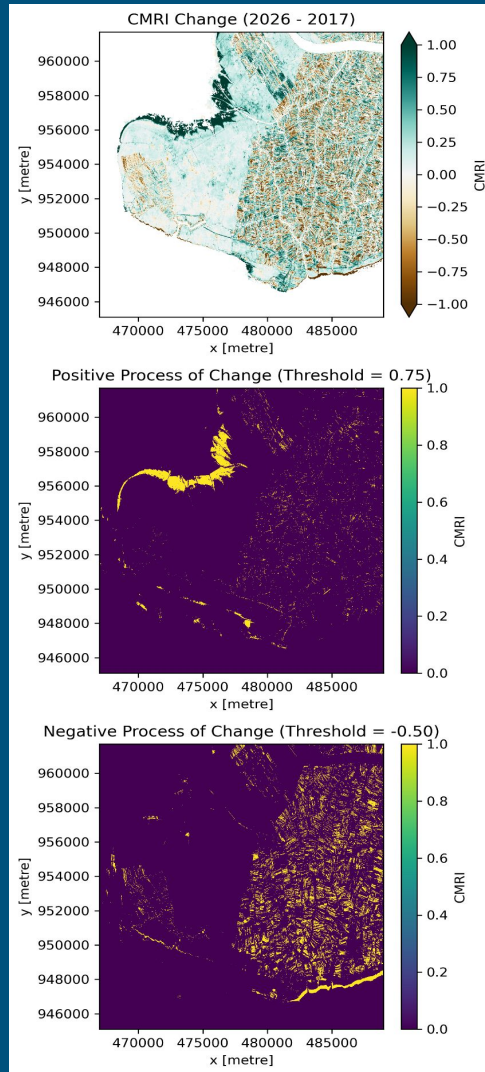
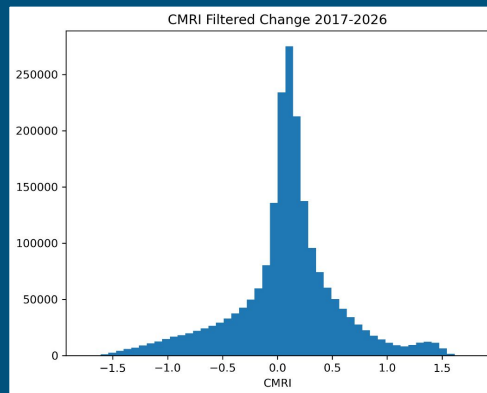
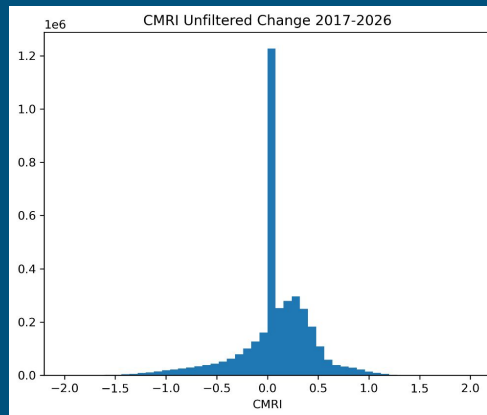
CMRI change std: 0.36



CMRI Threshold

Establishing threshold (similar process to NDVI)

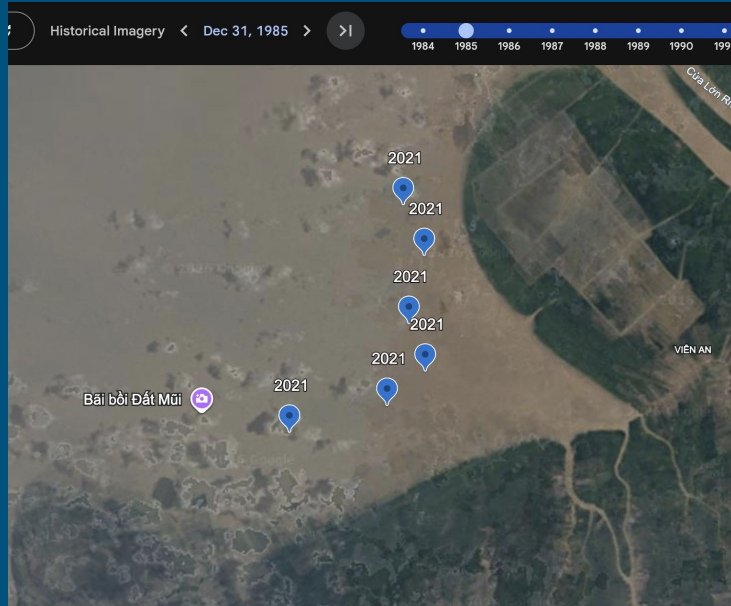
- Metrics below based off 0.1 change
- Vegetation Gain: 2.86%
- Vegetation Loss: 6.87%
- Stable: 40.57%



	Metric	Area (ha)
0	Total Raster Area 2017	36548.96
1	Mangrove Area 2017	18080.07
2	Total Raster Area 2026	36553.20
3	Mangrove Area 2026	17800.72
4	Mangrove Gain (2017-2026)	1043.99
5	Mangrove Loss (2017-2026)	2509.55

Imagery based classification

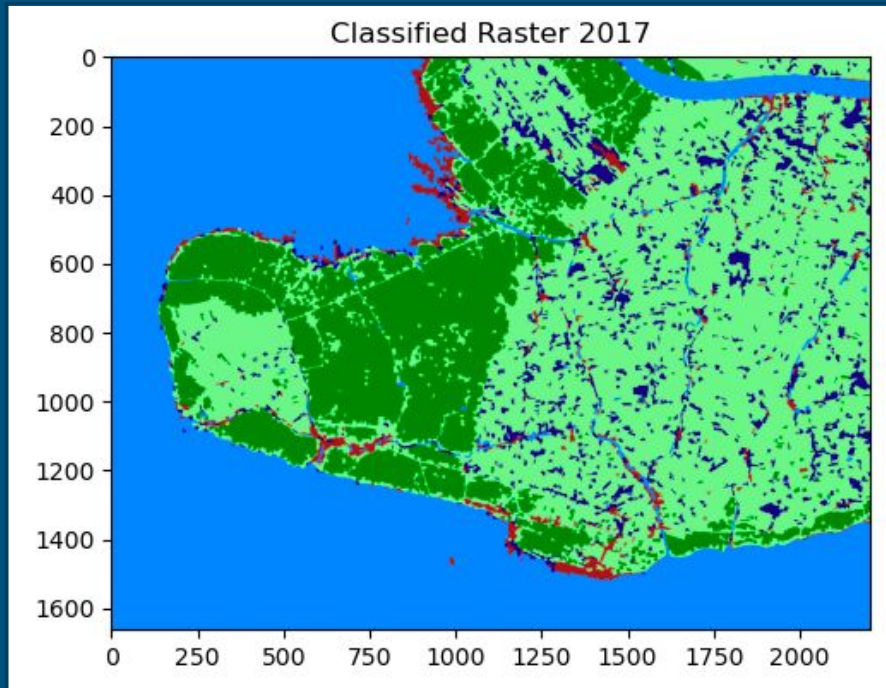
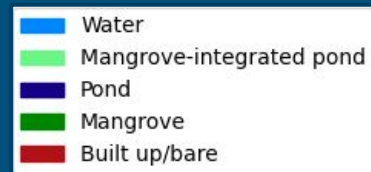
Using google earth to validate mangrove growth.



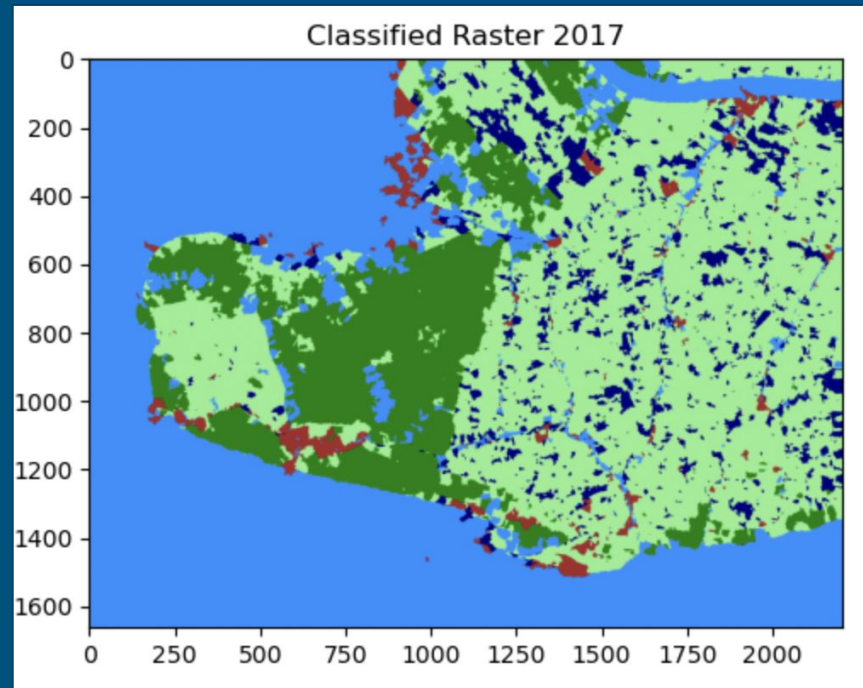
2017 vs 2021



Classified: 2017 image

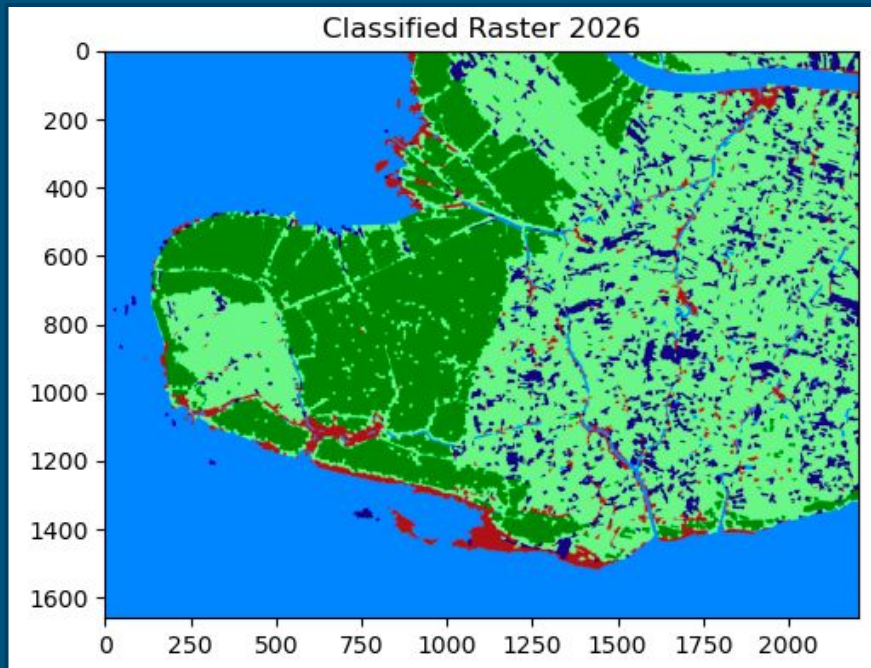
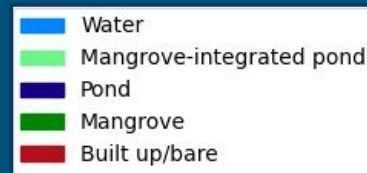


slic- n_segments=50000, compactness=0.1

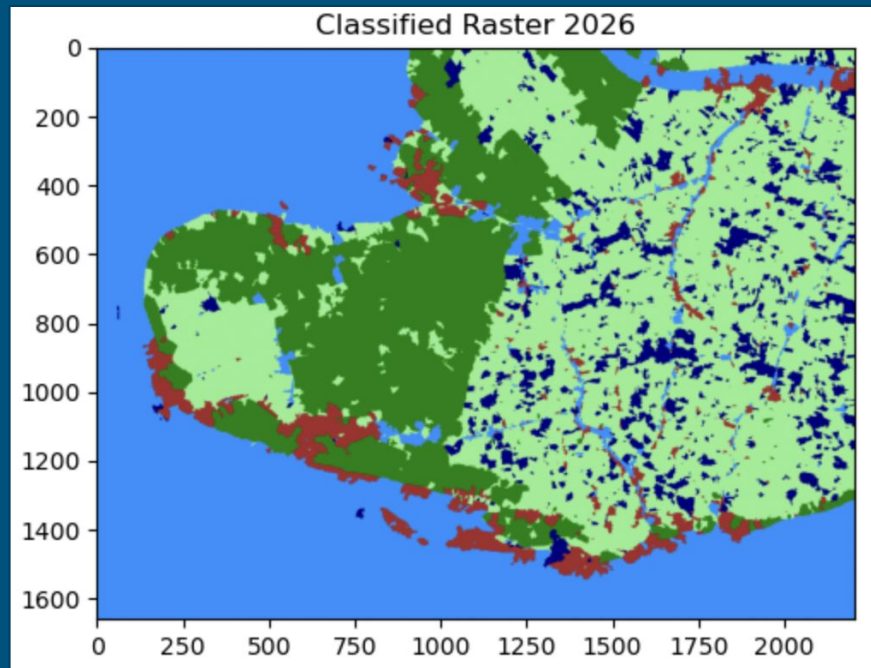


quickshift- ratio=.6, max_dist=4

Classified: 2026 image

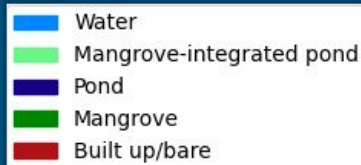


slic- n_segments=50000, compactness=0.1

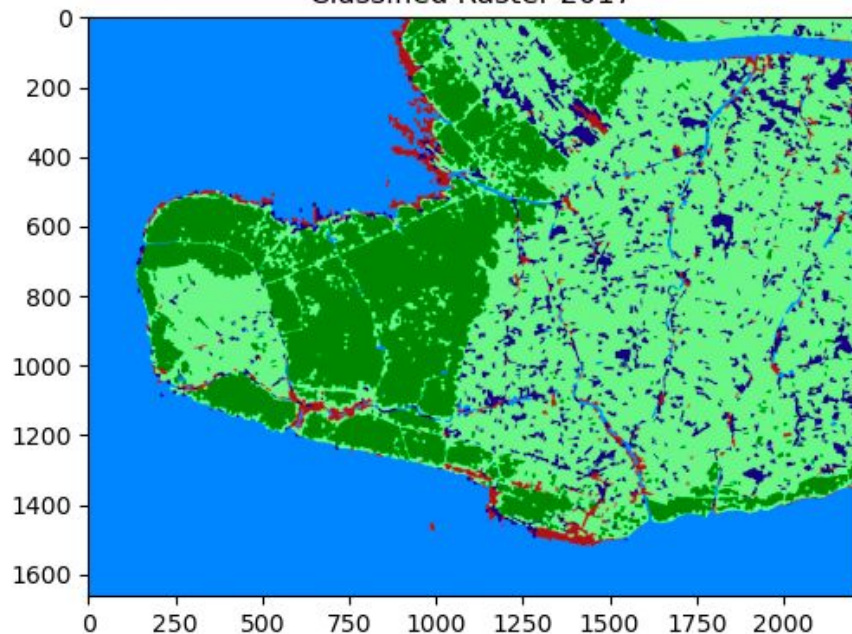


quickshift- ratio=.6, max_dist=4

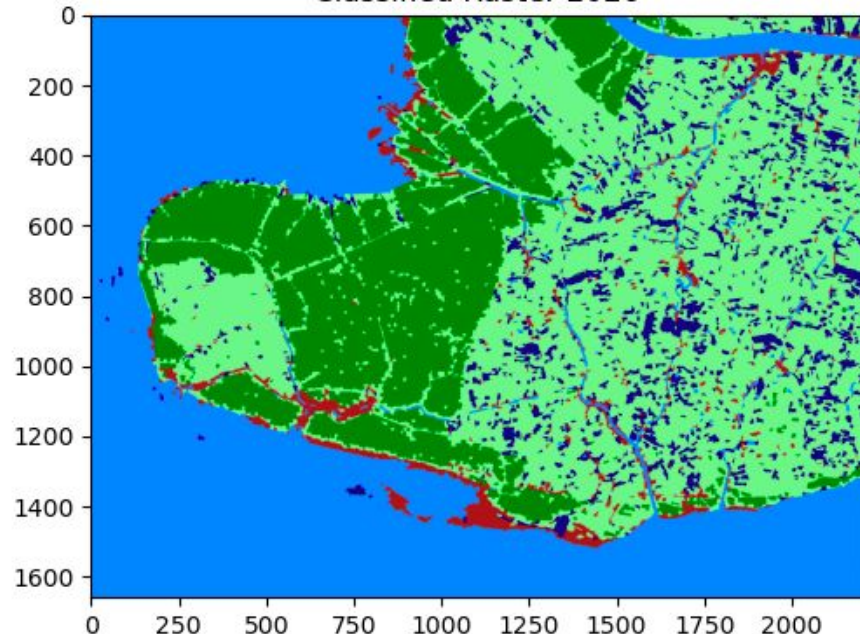
Change in classification between 2017 and 2026- what do we see?



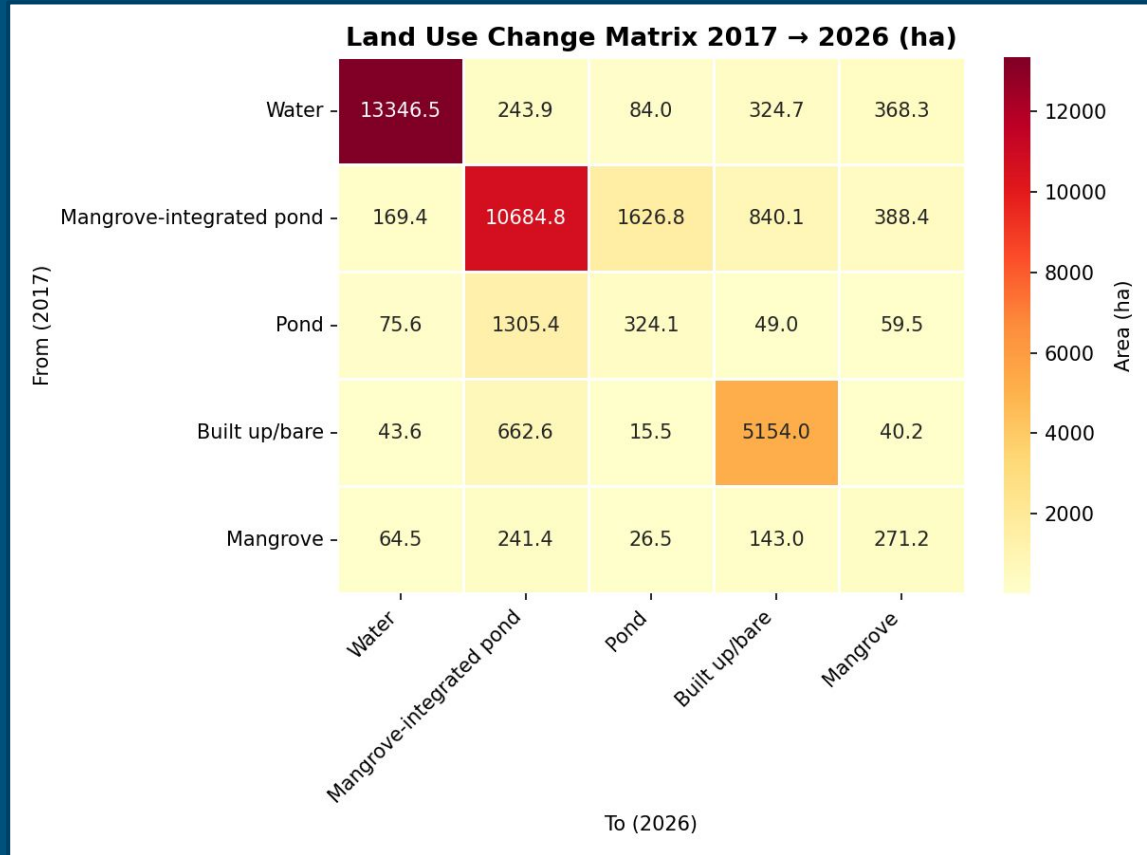
Classified Raster 2017



Classified Raster 2026



Land Use Change Matrix



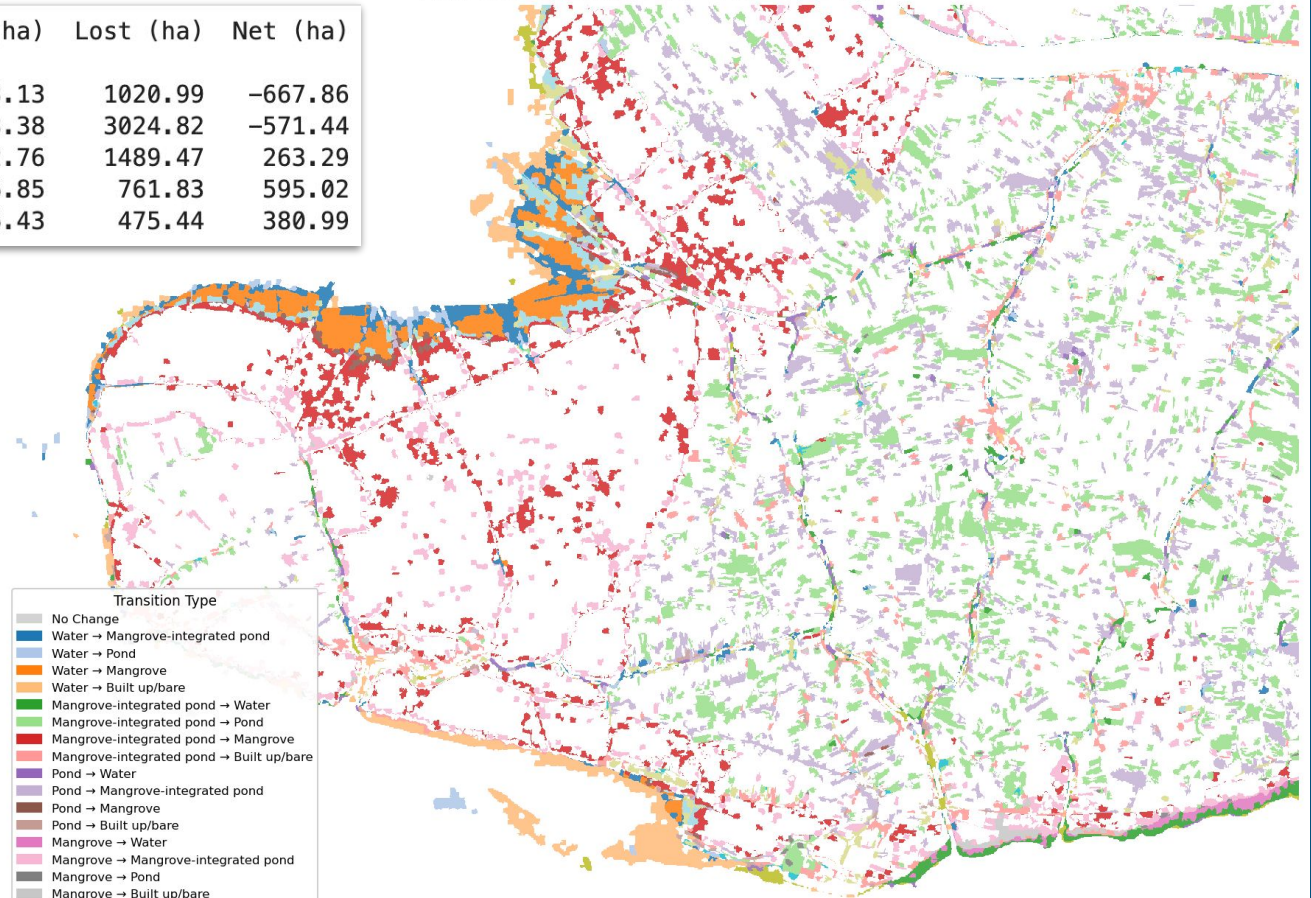
Land Use Transitions 2017 → 2026 — Ca Mau

Class	Gained (ha)	Lost (ha)	Net (ha)
Water	353.13	1020.99	-667.86
Mangrove-integrated pond	2453.38	3024.82	-571.44
Pond	1752.76	1489.47	263.29
Built up/bare	1356.85	761.83	595.02
Mangrove	856.43	475.44	380.99

Light green=
mangrove-integrated
pond -> pond

Light purple = pond ->
mangrove-integrated
pond

Orange = water ->
mangrove



Assessing accuracy (Method 1)

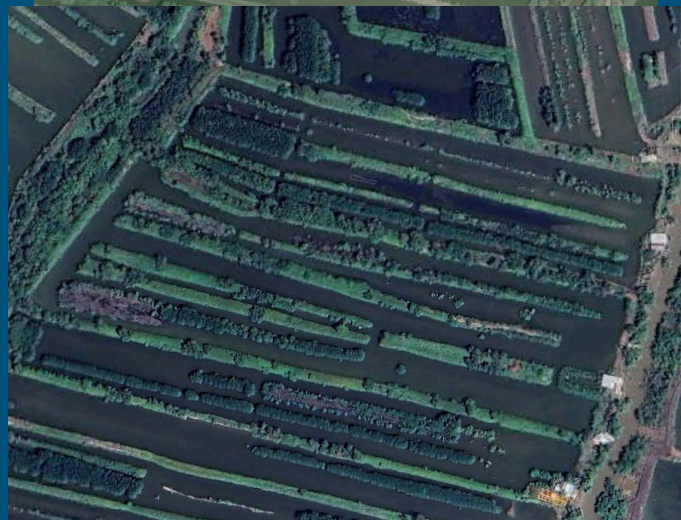
- 70% training data and 30% testing data
- Created confusion matrix between actual data and predicted data
- 84.6 % Accuracy

Confusion Matrix

	Actually Positive (1)	Actually Negative (0)
Predicted Positive (1)	True Positives (TPs)	False Positives (FPs)
Predicted Negative (0)	False Negatives (FNs)	True Negatives (TNs)

Predicted Class	User's Acc (Precision)	Producer's Acc (Recall)	True Positives	Total
Water 1094	0.9644	0.8873	1055	
Mangrove-integrated pond 10946	0.6737	0.9160	7374	
Pond 4037	0.8246	0.5423	3329	
Built up/bare 11131	0.9998	0.9529	11129	
Mangrove 1456	0.9636	0.8731	1403	
Overall Accuracy: 0.8474				





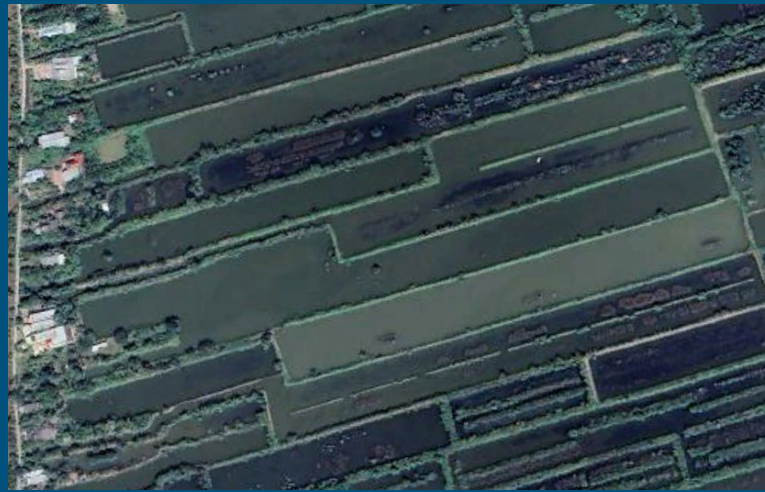
Conclusions/Challenges

- Overall, worked sort of well
- Switch from Landsat to Sentinel-2 (10 m resolution)
- Initially planned to clip image by elevation and distance to coast
 - Didn't end up using that threshold
- Wanted to use dry season (~ December - April)
 - 2016 dry season didn't have enough images to use
- Harmonization of new and old data
 - Sentinel changed their data parameters in 2022 to allow negative reflectance
 - Impacted CMRI Analysis

Conclusions/Challenges

- Cà Mau- nearly all vegetation in this tile will be mangroves
- Encountered both regrowth and afforestation (by humans)
- Patterns of planting and cutting within ponds
- Struggle with irregular pond shapes
 - Different models of silvofisheries and shrimp ponds
 - Potential obstacle to scaling up code to larger area
- Ambiguous line between mangrove integrated and non-mangrove integrated ponds

Intensive
ponds,
Bạc Liêu
Vietnam



Ponds w/o
mangrove
platforms,
Kiên Giang
Vietnam

Irregularly
shaped
ponds,
Manila Bay,
the
Philippines



Abandoned
ponds,
Mahakam
Delta,
Indonesia

Questions?

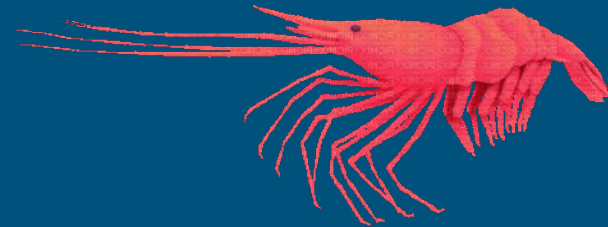
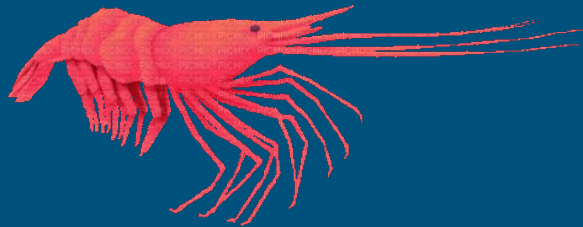


Image and Threshold References / Citations

- <https://www.sciencedirect.com/science/article/pii/S0048969722041729>
- <https://storymaps.arcgis.com/stories/47d57669202f4aa38739a294747ea28f>
- <https://pmc.ncbi.nlm.nih.gov/articles/PMC6174272>
- <https://www.mdpi.com/2072-4292/17/14/2512>

Further references can be found in our README file.